

Twitter Sentiment Analysis – Project Insights

Dataset Overview

- **Scale Analyzed:** 75,683 raw customer tweets parsed.
- **Dimensions:** 3 predictors (ID, Topic, Tweet) vs. 1 target (Sentiment).
- **Pipeline:** Automated preprocessing + Python EDA + Interactive Tableau Dashboard.

Sentiment Distribution

- **Negative Share:** 30.17% overall footprint (22,624 tweets).
- **Positive Share:** 27.91% overall footprint (20,932 tweets).
- **Neutral Share:** 24.53% overall footprint (18,393 tweets).
- **Irrelevant Share:** 17.40% overall footprint (13,047 tweets).

Brand Volume Leaders

- **Max Volume:** *League of Legends* leading with 2,414 total mentions.
- **Runner-Up Volume:** *Call of Duty* following at 2,407 total mentions.
- **Third Highest Volume:** *Madden NFL* tracking close at 2,406 total mentions

High-Risk Friction Points

- **Peak Brand Risk:** *Madden NFL* generated 1,712 purely negative tweets.
- **Secondary Risk:** Tom Clancy's *Rainbow Six* hit 1,124 negative tweets.
- **Corporate Friction:** Verizon logged 1,105 high-risk negative tweets.

Positive Engagement Drivers

- **Top Brand Champion:** *World of Warcraft* secured 744 positive tweets.
- **Secondary Champion:** *Apex Legends* pulled 644 positive tweets.
- **Community Wins:** *League of Legends* earned 624 positive tweets.

Data Cleaning Metrics

- **Null Filtering:** Detected and dropped 686 empty string records.
- **Data Integrity:** 74,996 fully valid rows preserved for model visualization.

Tableau Visual Engineering

- **Dashboard Footprint:** Fully interactive analysis mapping across 74,823 dynamic records.
- **Target Audience Focus:** Segmented KPI cards separating 3,757 negative user IDs vs. 3,472 positive user IDs.

- **Visual Scope:** Comprehensive coverage tracking **32** global entities simultaneously.
- **Cognitive Speed:** 4-color discrete maps (Positive: Green, Negative: Red, Neutral: Orange, Irrelevant: Blue) cutting insight delivery time down to under 3 seconds.

Business Insights & Strategic Takeaways

- **Prioritize Remediation:** Deploy instant product review sprints for brands experiencing extreme negative drops like Madden NFL
- **Replicate Success:** Extract community management frameworks from high-sentiment anchors like *World of Warcraft* to help lower-performing assets.
- **Real-Time Brand Safety:** Utilize dynamic cross-filtering to let marketing teams intercept sudden PR risks and protect digital brand equity.
- **Operational Efficiency:** Ensure data transformation pipelines filter out empty text noise before visualization to prevent misleading metric shifts.

Conclusion

- **Social media conversations** are **heavily shaped** by **industry sectors**, with **gaming driving the highest public participation**.
- **Cleaning empty data fields** and **standardizing category labels** is mandatory before trying to **build reliable visualization structures**.
- **Combining Python processing** with **interactive Tableau design** converts **chaotic stream data** into clear corporate strategy.